

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#define BUFFER_SIZE 5
```

```
int buffer[BUFFER_SIZE];
```

```
int in = 0, out = 0;
```

```
int mutex = 1;
```

```
int empty = BUFFER_SIZE;
```

```
int full = 0;
```

```
int item = 0;
```

```
void wait(int *s) {
```

```
    (*s)--;
```

```
}
```

```
void signal(int *s) {
```

```
    (*s)++;
```

```
}
```

```
void produce() {
```

```
    if (empty == 0) {
```

```
        printf("Buffer is full\n");
```

```
        return;
```

```
    }
```

```
wait(&empty);
```

```
wait(&mutex);
```

```
item++;
```

```
buffer[in] = item;
```

```
in = (in + 1) % BUFFER_SIZE;
```

```
printf("Producer has produced: item%d\n", item);
```

```
signal(&mutex);
```

```
signal(&full);
```

```
}
```

```
void consume() {
```

```
    if (full == 0) {
```

```
        printf("Buffer is empty\n");
```

```
        return;
```

```
    }
```

```
wait(&full);
```

```
wait(&mutex);
```

```
int consumed = buffer[out];
```

```
out = (out + 1) % BUFFER_SIZE;
```

```
printf("Consumer has consumed: item%d\n", consumed);
```

```
signal(&mutex);
```

```
signal(&empty);
```

```
}
```

```
int main() {  
  
    printf("1WA24CS216");  
  
    int choice;  
  
  
    while (1) {  
  
        printf("\nEnter your choice:\n");  
  
        printf("1. Producer\n2. Consumer\n3. Exit\n");  
  
        scanf("%d", &choice);  
  
  
        switch (choice) {  
  
            case 1:  
  
                produce();  
  
                break;  
  
  
            case 2:  
  
                consume();  
  
                break;  
  
  
            case 3:  
  
                exit(0);  
  
  
            default:  
  
                printf("Invalid choice\n");  
        }  
    }  
  
    return 0;  
}
```

1WA24CS216

Enter your choice:

1. Producer
2. Consumer
3. Exit

1

Producer has produced: item1

Enter your choice:

1. Producer
2. Consumer
3. Exit

1

Producer has produced: item2

Enter your choice:

1. Producer
2. Consumer
3. Exit

2

Consumer has consumed: item1

Enter your choice:

1. Producer
2. Consumer
3. Exit

2

Consumer has consumed: item2

Enter your choice:

1. Producer
2. Consumer
3. Exit

|